

Problem 2 Specification: Workforce Scheduling Optimization System SmarTask/Sisqual

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Abbreviations and Operational Codes

Acronym/Code	Definition
DL	Day Off (employee rest day)
DLF	Fixed Day Off (non-modifiable)
DLV	variable Day Off (reschedulable)
VAC	Vacation period
EnfD	Medical leave
DC-E	General absence
EG	Equipo Gestión (Management competency)
CAJ	Caja (Cashier competency)
ALM	Almacén (Warehouse competency)
KPI	Key Performance Indicator

Introduction

This document presents a formal specification of the workforce scheduling optimization problem, proposed by Sisqual. The goal is to formalize operational requirements, labor regulations, employee competency framework, and algorithmic challenges to be addressed by the SmarTask development team.

The problem specification is based on operational data, provided by the company through Excel files, and technical meetings, held on November 13 and December 5, 2025. This document serves as the internal technical reference for the project.

Problem Context and Operational Constraints

Store Operating Hours

The retail facility operates on a year-round basis with fixed hours:

- **Weekdays:** 8:30 AM – 10:00 PM

- **Weekends:** 11:00 AM – 9:00 PM
- **Public holidays:** No closures (365 days/year operation)

Input Data Structure

Operational data is provided in structured Excel files containing daily information on:

- Employee availability status
- Special markings (vacations, rest days, absences)
- Fixed and flexible shift assignments
- Employee competencies and proficiency levels

Scheduling Period

The planning period includes:

- 1 complete calendar month (target scheduling month)
- $x = 5$ additional days before the target month (configurable parameter)
- $y = 5$ additional days after the target month (configurable parameter)

These buffer periods incorporate pre-assigned shifts and rest days across month boundaries, preventing constraint violations from adjacent-day dependencies (e.g., maximum consecutive workdays, minimum rest period between shifts).

Employee Scheduling Codes System

Scheduling Codes

The system employs standardized notation in Excel files:

- **DL:** Day off, with two subtypes:
 - **DLF:** Fixed (immutable)
 - **DLV:** Variable (reschedulable within constraints)
- **VAC:** Vacation period
- **EnfD, DC-E:** Medical leave or general absence
- **Gray cells:** Contract gap (employee treated as newly hired upon rehiring)
- **Red markings:** Fixed shifts (immutable by algorithm)
- **Flexible shifts:** Fixed duration (4h, 5h, 7h, 8h) with algorithm-determined start/end times

Unavailability Scheduling Codes

DLF, VAC, EnfD, and DC-E represent absolute unavailability and are treated identically: employees with these markings are excluded from scheduling for the corresponding periods.

Employee Competency Framework

Competency Types

The system recognizes three competency categories:

- **EG (Management):** Supervisory and management functions
- **CAJ (Cashier):** Point-of-sale operations
- **ALM (Warehouse):** Inventory management

Proficiency Levels

Proficiency levels are inversely scaled: level 1 denotes maximum proficiency ($\ell_{\min} = 1$). Employees may possess multiple competencies at different proficiency levels (e.g., EG-3, CAJ-1).

Scheduling Requirements and Constraints

The system makes a distinction between hard constraints (legally required, cannot be violated) and soft constraints (operational preferences guiding optimization).

Hard Constraints (legally bonded)

1. **Maximum consecutive workdays:** No more than $k = 5$ consecutive days without rest
2. **Minimum required rest between shifts:** At least $h = 11$ hours between shifts
3. **Mandatory rest days:** It must be assured the defined minimum rest days per employee (set on a weekly or monthly basis, depending on the applicable jurisdiction).

Soft Constraints (contract binding)

1. **Temporal granularity:** Shift times in Δt intervals ($\Delta t = 30$ min, configurable)
2. **Rest exchanges within the same week:** DLV may only be rescheduled in the same calendar week
3. **Part-time employees:** Part-time employees ($d = 2$ or 3 days/week) can not switch their days off.
4. **Mandatory Sunday rest:** At least $s = 1$ Sunday off per month

Week Definition

By default, a week spans from **Monday to Sunday**. The system allows configuration of the first day via parameter `week_start_day` $\in \{\text{Monday, Sunday, } \dots\}$ in a JSON file, with Monday as default.

Constraint Validation

When client-imposed rules violate labor legislation or are logically inconsistent, the system flags the situation as *illegal*. Neither the algorithm nor development team assumes responsibility for resulting schedules in such cases.

Alarm System and KPIs

Alarms system define staffing requirements over 30-minute intervals. The alarms indicate the minimum requirement for employees with specific skills (EG, CAJ, or ALM) during specific periods.

KPI Levels

Three staffing levels (measured in *employees per time slot*) are defined per task and time interval:

- **Minimum** ($M_{\min}$): Minimum number of employees/time slot required to ensure the basic execution of a task, within a given time interval. If the minimum requirement is not met, the task is considered incomplete, raising a red flag.
- **Ideal** (M_{ideal}): Target level of employees/time slot that represents the balance between operational efficiency, service quality, and resource management. It corresponds to the recommended workload to perform the task in a stable manner, avoiding overload or underload.
- **Estimated** (M_{est}): Predicted value of employees/time slot, based on historical standards or operational estimates, used as supplementary guidance in planning. It is not mandatory, but it helps to adjust the distribution of resources according to expected demand.

Maximum alarm threshold: $M_{\max} = \max (M_{\min}, M_{\text{ideal}}, M_{\text{est}})$.

Task Minimum Interpretation

The minimums defined in the alarms refer to requirements per task and per 30-minute interval (time slot), expressed as employee per time slot, and not to the total number of employees present in the store. The sum of the minimums across the day represents the total required workload (measured in employees per time slot), and not the number of employees present at any given time. Therefore, the minimums are not defined “per employee,” but per task per time slot, allowing the algorithm to distribute the required workload across employees and reassign tasks at each 30-minute time slot, as needed.

Resource Insufficiency

When the minimum tasks exceed the available employees, it is possible that, in a given 30-minute period, the sum of the minimum tasks exceeds the number of employees available during that period. In these cases:

- The algorithm must apply the alarm priority hierarchy, ensuring critical tasks are performed first.
- Tasks whose minimum cannot be fulfilled must be marked in red.

This behavior does not indicate an error in the data, but rather the need to prioritize tasks when resources are insufficient.

Employee Flexibility

An employee can perform different tasks throughout the day, changing tasks in 30-minute blocks. Thus, they can contribute to multiple minimums throughout the day, as long as they do not perform different tasks simultaneously. This allows the total workload to be completed with a reduced number of employees, provided that there is adequate time distribution.

Overall, the minimums per task represent requirements expressed in employees per time slot. An employee can contribute to different tasks throughout the day, and when the minimums cannot be met simultaneously, alarm priorities come into play and tasks that are not guaranteed are flagged.

Alarm Application Levels

Alarms can be applied at different organizational levels:

- **By Staff:** Includes all store employees. An alarm applied at the staff level considers any available employee, regardless of team or function.
- **By Team:** Corresponds to a specific operational group within the organization (e.g., CAJ). An alarm per team considers only the employees of that team.
- **By Type of Team:** It combines several similar teams within the same functional category (e.g., all teams associated with the cash register). An alarm by team type allows several teams to contribute to the same requirement.
- **By Task:** Defines a specific operational requirement encoded by the alarm, such as CAJAN=1 or RESP_ALMACEN.N $\geq$ 1, which may include specific requirements for skills, level, and amount.

Alarm Priority Hierarchy

The alarm priority hierarchy follows the order below:

1. RESP ALMACEN N $\geq$ 1 — Warehouse supervisor, level $\geq$ 1
2. RESP - EQUIPO GESTION N=1 — Management supervisor, level 1
3. CAJA N=1 — Cashier, level 1
4. RESP - EQUIPO GESTION N=2 — Management supervisor, level 2
5. CAJA N=2 — Cashier, level 2
6. RESP - EQUIPO GESTION N=3 — Management supervisor, level 3
7. RESP - EQUIPO GESTION N $\geq$ 4 — Management supervisor, level $\geq$ 4
8. CAJA N=3 — Cashier, level 3
9. EQUIPA Empleados N $\geq$ 1 — General employee, level $\geq$ 1

Resource Allocation Rule

Fundamental principle: Resources should not be assigned to meet the ideal value of one task if the minimum value of another task cannot be met. This rule is particularly relevant, because there are employees with multiple skills (e.g., management and cashier), and their assignment must be managed efficiently.

Challenge 1: Optimization of Time Off Swaps

Objective: Develop an algorithm that minimizes fixed days off swaps while maximizing service coverage.

Features:

- Establishing fixed days off improves employees' quality of life.
- The algorithm should only swap days off when necessary to improve alarm coverage.
- Part-time employees (2-3 days/week) cannot have their days off swapped.
- Red markings (fixed schedules or days off) cannot be changed.
- Day off exchanges can only be arranged within the same week.

Challenge 2: Sequential Allocation by Proficiency Level

Objective: Develop an algorithm that assigns employees to teams following a competency-based priority order while enforcing resource allocation constraints.

Features:

- The staffing order may vary depending on the day of the week and the need to meet KPIs.
- Must respect the defined alarm priorities.
- The algorithm must follow resource allocation rules, ensuring that the ideal value of one task is not used if it compromises the minimum value of another task.

Algorithmic Challenge 3: Multi-Competency Employee Allocation

Objective: Develop an algorithm that optimally decides the best allocation for an employee when they have multiple skills. The algorithm must consider both competency type and proficiency level for each employee.

Example: There is one available employee available with CAJ-1 and EG-3 skills. Should the employee be assigned to the cashier role (level 1) or to management (level 3)?

Note: This is a novel contribution. There is no similar solution on the market.

Factors that should be considered:

- Alarm priorities
- Employee proficiency levels (e.g., EG-1, CAJ-2)
- KPI fulfillment (minimum, ideal, estimated)
- Long-term impact on subsequent periods

Challenge 4: Break Management

Objective: Support two break handling modes via parameter `break_mode` $\in \{\text{explicit}, \text{implicit}\}$:

1. **Explicit mode:** Include breaks for meals and rest
 - Higher complexity
 - Breaks vary according to the schedule and duration of the shift
 - Depends on the country's legislation
2. **Implicit mode:** Breaks included in shift period
 - Lower complexity

Proposed Extensions

DLV Movement Policy

Proposed extension: integrate DLV rescheduling into the core algorithm via a configurable parameter `rest_movement_policy` $\in \{\text{within_week}, \text{within_month}, \text{fixed}\}$:

- `within_week`: rescheduling allowed only within the same calendar week
- `within_month`: rescheduling allowed only within the same calendar month
- `fixed`: no rescheduling (default)

This enables systematic evaluation of alternative rest-day allocation policies. The system should support KPI computation with and without DLV rescheduling, in order to quantify the policy impact on schedule outcomes.

Schedule Construction Models: Shifts vs. Time Intervals

The system should support two schedule construction models:

1. a shift-based model (e.g., M/T/N)
2. a time-interval model with 30-minute granularity.

This allows the algorithm to be adapted to different operational settings and required levels of temporal detail. When using the shift-based model, a consistent mapping to the underlying time-interval representation must be defined for constraint and KPI evaluation.